

Georg von Westphalen

MedKayfab: Oncology Wrestling!

The Hall of Malignant Mirrors (Uncle FrizzleBob's Krazy Kayfabizzaro Komik Kards)



BLÖDSINN!
Stay fluffy!

GORDON THE GORILLA (Self-Sufficiency)

(Autocrine Loop /
ERBB2 (HER2))



POWER: The Infinite Ego. Gordon screams at himself to grow, ignoring the body's "Stop" signals.

LORE: Mutations in EGFR or RAS keep the growth engine floored even when the key is out of the ignition.

What the FLUFF?!

THE P53 POLICE BULLDOG (Tumor Suppressor)

(Li-Fraumeni Syndrome/
TP53 Mutation)



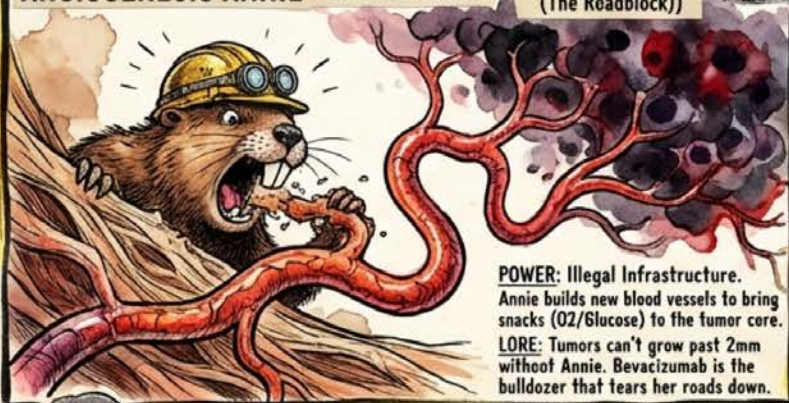
POWER: The Broken Badge. Normally stops the cycle or hits "Self-Destruct." When the badge breaks, anything goes.

LORE: The "Guardian of the Genome." If P53 is down, the cell ignores DNA damage and keeps wrestling.

SYSTEM FAILURE!

ANGIOGENESIS ANNIE (The Road Builder)

(VEGF / Bevacizumab
(The Roadblock))



POWER: Illegal Infrastructure. Annie builds new blood vessels to bring snacks (O2/Glucose) to the tumor core.

LORE: Tumors can't grow past 2mm without Annie. Bevacizumab is the bulldozer that tears her roads down.

TELOMERASE TERRY (The Eternal Wrestler)

(TERT Promoter Mutations)



POWER: Immortal Ends. Terry puts caps back on his shoelaces (Chromosomes) so they never fray.

LORE: Normal cells have a Hayflick Limit. Terry ignores it. He doesn't age; he just grows.

EPIDEMIC
OF
IGNORANCE

NEVER
SAY DIE!

MEDICAL MISFITS: ROUND 2.

METASTASIS MAX (The Wanderer)

E-Cadherin ↓ /
Snail & Twist ↑



POWER: Basement Membrane Breach. Max dissolves the ropes and leaves the ring to invade the parking lot.

LORE: EMT (Epithelial-Mesenchymal Transition). Max loses his anchors (Cadherins) and enters the bloodstream highway.

Rb THE GATEKEEPER (The G1/S Referee)

Retinoblastoma



POWER: The Open Gate. If phosphorylated, he lets the cell cycle start. If mutated, the gate is always open.

LORE: Master regulator of the G1 to S phase transition.

KayfaBINGO!

BCL-2 BOB (The Anti-Death Shield)

Overexpressed in t(14;18)



POWER: Apoptosis Block. He prevents the mitochondria from releasing the "Apoptosis Grenade."

LORE: Overexpressed in Follicular Lymphoma. Cells that should die are kept artificially alive.

WARBURG WALLY (The Glycolysis Junkie)

High FDG Uptake (PET Scan)



POWER: Aerobic Fermentation. Even with plenty of oxygen, Wally insists on fermenting his sugar.

LORE: This massive, inefficient glucose appetite makes tumors glow on PET scans.

KayfaBINGO!

PD-L1 PETER (The Invisibility Cloak) Immune Checkpoint Target



POWER: The 'Self' Mack. Peter wears a mack that tells T-Cell Bouncers "I belong here."
LORE: Checkpoint inhibitors rip this mack off so the immune system can attack.

RAS THE ROGUE (The Broken Switch) Permanent 'ON' State



POWER: Untoppable Relay. Once flipped to 'ON' by a growth factor, he stays there forever.
LORE: A major downstream signaling protein. If it can't turn off, the cell divides endlessly.

MYC THE MEGAPHONE (The Amplifier) t(8;14)



POWER: Deafening Roar. MYC amplifies every single growth signal into a catastrophic shockwave.
LORE: A transcription factor overdrive, famously driving Burkitt Lymphoma.

BRCA BRENDA (The DNA Repairwoman) Double-Strand Break Repair Failure



POWER: Unfixed Mayhem. When Brenda is on strike (mutation), double-strand breaks go unfixed.
LORE: High risk for breast and ovarian cancer mayhem.

KayfaBIZARRO!

Sergio BIZARRO!

MEDICAL MISFITS: ROUND 3 — THE GENE GANG

APC THE TRASH COMPACTOR

FAP / Beta-Catenin Buildup



POWER: Compactor Failure. Usually degrades Beta-Catenin. Without APC, it builds up.
LORE: Familial Adenomatous Polyposis (FAP) forces the colon to grow 1000s of polyps.

VHL THE SNORKEL

Renal Cell Ca



POWER: Phantom Suffocation. Normally detects oxygen. If mutated, the cell thinks it's suffocating.
LORE: The hypoxic panic summons Angiogenesis Annie to build massive vessels.

NF-1 THE BRAKEMAN

Neurofibromatosis Type 1



POWER: The Snapped Reins. He normally keeps RAS in check.
LORE: Without him, nerves grow uncontrollably into tumors.

ALL-IN ALICE (Acute Lymphoblastic Leukemia)

TdT+ / Child < 15yo
 t(12;21) Good / t(9;22) Bad



POWER: The Schoolyard Takeover. Alice fills the marrow with 'Blasts' until there's no room for normal blood.
LORE: Most common childhood cancer. Hides in the 'Sanctuaries' (CNS and Testes).

Diagnosis?!

Diagnosis?!

MEDICAL MISFITS: ROUND 4 - THE BLOOD BLASTERS

AML ARCHIE (Acute Myeloid Leukemia)

Auer Rods / MPO+
t(15;17) (APL)



POWER: The Needle Strike. Archie's Auer Rods trigger DIC.
LORE: APL is a bleeding emergency. Treat with ATRA (Vitamin A) to force blasts to grow up.

CLL LARRY (Chronic Lymphocytic Leukemia)

Smudge Cells
CD5+



POWER: The Fragile Crowd. Larry lives forever but is easily crushed during a blood smear.
LORE: Elderly patients. "Wait and Watch." Can transform into aggressive large B-cell lymphoma.

CML CHARLIE (Chronic Myeloid Leukemia)

Philadelphia Chromosome
t(9;22) / BCR-ABL



POWER: The Tyrosine Rocket. Charlie is powered by a constitutively active fuse.
LORE: Treat with Imatinib (The Fuse-Cutter). High basophils, low LAP score.

BURKITT'S BARRY (The Starry Night)

t(8;14)
c-myc / EBV



POWER: Jawbreaker Blast. Extremely fast doubling time driven by MYC.
LORE: "Starry Sky" histology is caused by macrophages eating the massive debris field of dying cells

KayfaBOGGLE!



KayfaBOGGLE!



MEDICAL MISFITS: ROUND 5 - THE LYMPHOMA LORDS

HODGKIN'S HARRY

Reed-Sternberg "Owl Eyes"
CD15+ / CD30+



POWER: The Reactive Mob. He's a lonely malignant B-cell hiding in a massive crowd of reactive inflammation.
LORE: The tumor is mostly healthy cells; the Reed-Sternberg cells are rare giants.

LYMPH-MANIA!

HUMBUG!

FOLLICULAR FIONA

t(14;18)
BCL-2 Overloader



POWER: Painless Fluctuation. Her nodes wax and wane over months without causing pain.
LORE: Driven entirely by the BCL-2 Anti-Death Shield. Hard to cure, very slow-moving.

NODES GONE WILD!

MANTLE CELL MANNY

t(11;14) /
Cyclin D1 driver

POWER: Cell Cycle Smash. A very aggressive bouncer that overdrives the G1/S transition.

LORE: Poor prognosis, older males. Highly aggressive non-Hodgkin lymphoma.



NODES GONE WILD!

BIO-HAZARD BEATS!

MARGINAL ZONE MIKE

MALToma /
Associated with Chronic Inflammation

POWER: The Inflammatory Squatter. Mike only sets up shop where chronic inflammation has been burning for years.

LORE: Eradicating the *H. pylori* infection can sometimes cure the gastric MALToma entirely.



ANARCHY IN THE GLANDS!

MEDICAL MISFITS: ROUND 6 - THE MARROW MARAUDERS

MULTIPLE MYELOMA MONTY (The Bone-Cracker)

M-Spike (IgG/IgA) / CRAB Symptoms



POWER: Osteolytic Sabotage. Dissolves the skeleton.

LORE: "CRAB": Calcium ↑, Renal failure, Anemia, Bone lesions. Light chains in urine.

WALDENSTRÖM WALLY

IgM M-Spike / Hyperviscosity



POWER: The Giant Antibody. Makes the blood thick like sludge.

LORE: Causes visual and bleeding issues. Crucially: No bone lesions!

POLYCYTHEMIA PAUL

JAK2 Mutation



POWER: Erythroid Overload. Paul makes far too many red blood cells.

LORE: Classic presentation: Pruritus (severe itching) after a hot shower.

ESSENTIAL TOBY (Thrombocythemia)

JAK2 / Massive Platelet Counts



POWER: The Clot-Bleed Paradox. Toby makes so many defective platelets you bleed and clot at the same time.

LORE: A disorder of megakaryocytes. Diagnosis of exclusion.

BLOOD THICK ENOUGH TO CHEW!



WEB OF CHAOS!

Stay fluffy!



BONEZ R 4 BREAKIN'!

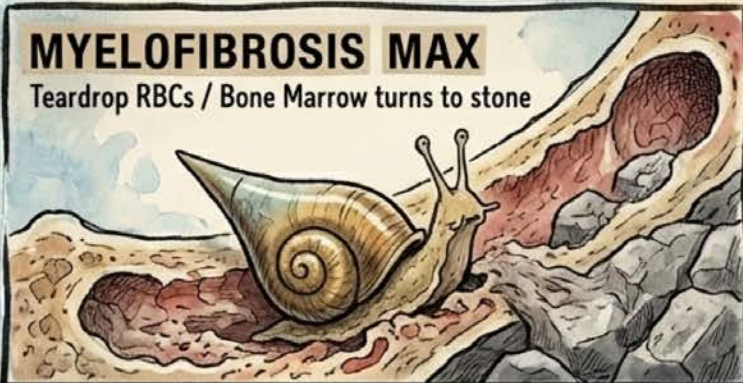


↖ JAK2 THE FUTURE! ↗

MEDICAL MISFITS: ROUND 7 - THE MARROW MINERS & NEURO NINJAS

MYELOFIBROSIS MAX

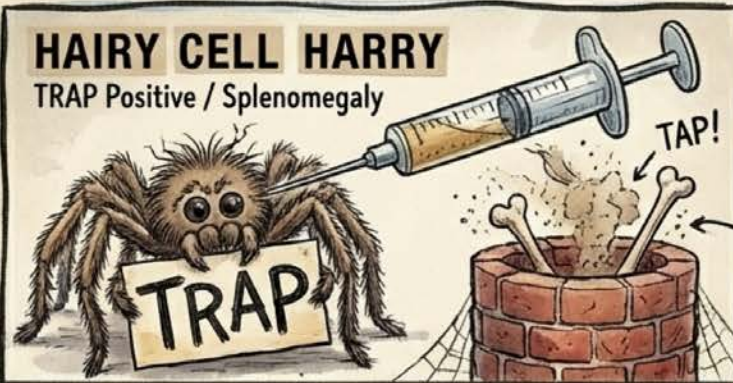
Teardrop RBCs / Bone Marrow turns to stone



POWER: Fibrotic Eviction. Max replaces the marrow factory with useless scar tissue.
LORE: 'Teardrop' RBCs happen because they have to squeeze out of the fibrotic marrow.

HAIRY CELL HARRY

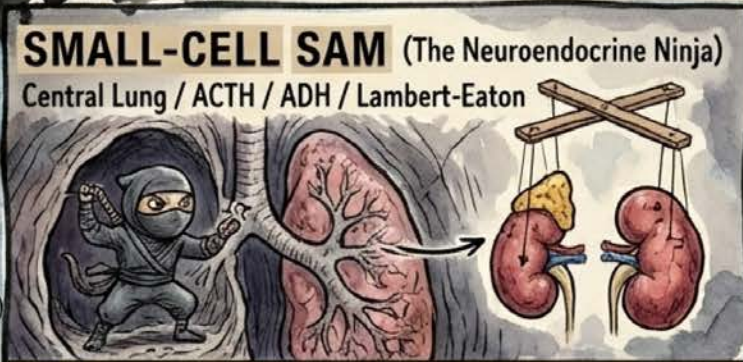
TRAP Positive / Splenomegaly



POWER: The Velvet Web. Harry gets tangled in the marrow and spleen.
LORE: Causes a 'dry tap' on marrow aspiration. Highly responsive to Cladribine.

SMALL-CELL SAM (The Neuroendocrine Ninja)

Central Lung / ACTH / ADH / Lambert-Eaton



POWER: Hormonal Sabotage. Sam spells that make you retain water (SIADH) or look like a moon-face (Cushing).

LORE: Very aggressive. Too small for surgery, but Sam hates Chemo/Radiation.

RENAL CELL REX (The Polycythemia King)

VHL Mutation / Clear Cell / t(3;?)



POWER: The EPO Overdose. Rex makes too much Erythropoietin, turning blood into sludge.

LORE: Loves the renal vein. If he reaches the IVC, he climbs to the heart like a malignant ladder.



KayfaBONGO!

WEBHEAD!

SPLEEN SQUEEZE!

← IT'S A TRAP!

BLOOD SLUDGE!



RENAL THRONE!

EPO KING!

TOO SMALL!

← NINJA ATTACK! →

KING REX RULES! →

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MEDICAL MISFITS: ROUND 8 - THE ORGAN OUTLAWS

PANCREATIC PETE (The Silent Assassin)

CA 19-9 / Courvoisier's Law
(Painless Jaundice)



POWER: Trousseau's Sign. Pete makes blood clot in random places (Migratory Thrombophlebitis).
LORE: Usually found too late. If Pete is in the head of the pancreas, look for the "Double Duct" sign.

SQUAMOUS SID

PTHrP producer
Central lung cancer



POWER: Calcium Crush. Sid secretes fake Parathyroid Hormone, leaching calcium.
LORE: Causes severe Hypercalcemia (Stones, Moans, Groans).

ADENO ADDIE

Peripheral lung cancer
Non-smokers



POWER: Glandular Architecture. Addie builds mucin-producing factories in the periphery.
LORE: The most common lung cancer in non-smokers.

COLON CAL (The Apple-Core)

Left-sided (Obstruction)
vs. Right-sided (Bleeding)



POWER: The Stricture Choke. Left-sided lesions obstruct and cause "pencil stools".
LORE: Right-sided lesions bleed silently, causing iron-deficiency anemia.

What the
FLUFF?!



MEDICAL MISFITS: ROUND 9 - THE CELLULAR CHAOS CREW.

CHAOS, TUMORS & TERRIFYING TOADS!

HEPATOCELLULAR HERB

Alpha-Fetoprotein (AFP)
Hepatitis B/C



POWER: The Scar-Dweller. Herb sets up shop in livers destroyed by chronic Hep B/C or alcohol.
LORE: Reactivates AFP, a fetal marker. Spreads hematogenously via the hepatic vein.

CELLULAR SABOTAGE!

TERATOMA TERRY (The 'Monster' Tumor)

3 Germ Layers



POWER: The Flesh Lottery. Terry spontaneously grows any tissue in the body.
LORE: If it's immature, it's malignant. If mature, generally benign (except in males).

TUMOR
TIME!

GLIOBLASTOMA GUS (The Butterfly Glioma)

Crosses Corpus Callosum
Pseudopalisading necrosis



POWER: Carpus Crossing. Gus ignores the brain's midline barrier, spreading wings of necrosis.
LORE: Histology shows cells lining up (palisading) around dead tissue.

TUMOR TIME!

SEMINOMA SAM

Fried-egg cells
↑ ALP /
Radiosensitive



POWER: The Clear Clene. Sam builds uniform sheets of perfect fried-egg cells.
LORE: Excellent prognosis because Sam is highly susceptible to radiation.

ANARCHY
IN THE OR!

Diagnosis?!
WTF?!



Spots won't
go away!
HELP!

MEDICAL MISFITS: ROUND 10 - THE REPRODUCTIVE & ABDOMINAL ANARCHISTS

PREGNANCY, PAIN & PERITONEAL PERIL!

CHORIOCARCINOMA CORY ↑β-HCG / Hematogenous Spread



POWER: The Blood Cannon. Cory fires 'cannonball metastases' straight into the pulmonary blood stream. | **LORE:** Highly aggressive malignancy of trophoblastic tissue.

BRCA BRENDAS OVARIAN COUSIN

CA-125 / Serous Cystadenocarcinoma



POWER: The Silent Swell. She fills the abdomen with ascites before anyone notices. | **LORE:** Psammoma bodies are laminated, calcified concretions seen on histology.

PHEOCHROMOCYTOMA PHIL (The 10% Tumor)

The 5 P's / Catecholamines



PRESSURE! PAIN!
PERSPIRATION!
PALPITATIONS!
PALLOR!

POWER: The Catecholamine Storm. Phil dumps adrenaline, maxing out blood pressure. | **LORE:** Pressure, Pain, Perspiration, Palpitations, Pallor.

GASTRIC GARY

Virchow's Node / Sister Mary Joseph sign



POWER: The Lymphatic Telegraph. Gary signals his presence in the stomach by enlarging distant, specific lymph nodes. | **LORE:** A swollen left supraclavicular node is a massive red flag for abdominal malignancy.

ANARCHY IN THE ABDOMEN!

WHAT THE FLUFF?!

PSAMMOMA PARTY!

LYMPH NODE SIGNAL!

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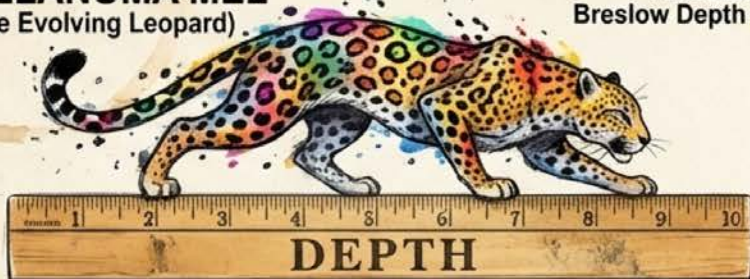
HEAVY METAL CHEMO & TOXIC TERRORS!

DOSE, DESTRUCTION & CELLULAR SABOTAGE!



MELANOMA MEL (The Evolving Leopard)

ABCDE /
Breslow Depth



POWER: The Deep Dive.
Depth (Breslow) is everything.

LORE: Asymmetry, Border, Color, Diameter,
Evolving. (Bridges Solid to Pharma).

CISPLATIN SID (The Platinum Hammer)

Nephrotoxicity /
Ototoxicity



POWER: DNA Cross-Linker.
Sid hammers the DNA strands together
so they can't unzip for replication.

LORE: He breaks the kidneys and
makes you deaf. Give Amifostine to
shield the kidneys.

DOXORUBICIN DOT (The Red Devil)

Cardiotoxicity /
Dilated Cardiomyopathy



POWER: The Heart-Breaker. Dot is
beautiful but she creates free radicals
that eat the heart muscle.

LORE: Use Dexrazoxane (The Shield)
to prevent Dot from trashing the pump.

VINCRIStINE VINNY (The Nerve-Stripper)

Peripheral Neuropathy /
Microtubule Inhibitor



POWER: Spindle Sabotage. Vinny
stops cell division at M-phase by
freezing the pipes.

LORE: Side effect: 'Foot drop' and
'Glove-and-stocking' numbness. Don't let
Vinny touch the CNS (Fatal if intrathecal).

NO FUTURE
FOR TUMORS!

ANARCHY
IN THE VEINS!



CYCLOPHOSPHAMIDE CY (The Bladder-Burner)

Hemorrhagic Cystitis / Acrolein



POWER: The Alkylating Agent. Cy cross-links DNA but leaves a toxic trash called Acrolein in the bladder.
LORE: Always deploy Mesna to neutralize the Acrolein trash before the bladder starts bleeding.

ANARCHY IN THE BLADDER!

5-FU FRED (The Pyrimidine Pirate)

Inhibits Thymidylate Synthase



POWER: The Fake Nucleotide. Fred jams the DNA printer permanently.
LORE: Classic side effect is Hand-foot syndrome.

METHOTREXATE MATT (The Folate Thief)

Inhibits Dihydrofolate Reductase



POWER: Synthesis Starvation. Matt steals the active folate needed to print new DNA.
LORE: Rescue the normal cells with Leucovorin (folinic acid bypass).

FOLATE FRENZY!

IMATINIB IZZY

The BCR-ABL Fuse-Cutter



POWER: Targeted Neutralization. Izzy shuts down the specific mutant tyrosine kinase.
LORE: Targeted therapy for CML. Turned a fatal disease into a manageable one.

CHEMO RIOT!

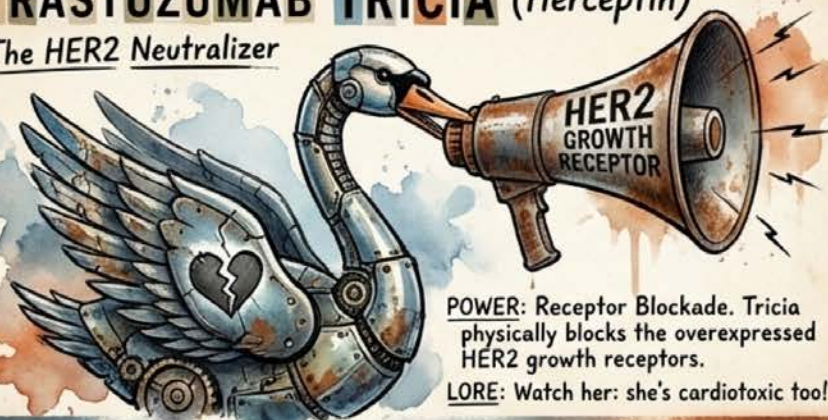
NO FUTURE FOR TUMORS!



NO FUTURE FOR TUMORS!

TRASTUZUMAB TRICIA (Herceptin)

The HER2 Neutralizer



POWER: Receptor Blockade. Tricia physically blocks the overexpressed HER2 growth receptors.

LORE: Watch her: she's cardiotoxic too!

PEMBROLIZUMAB PENNY (The Checkpoint Sniper)

PD-1 Inhibitor

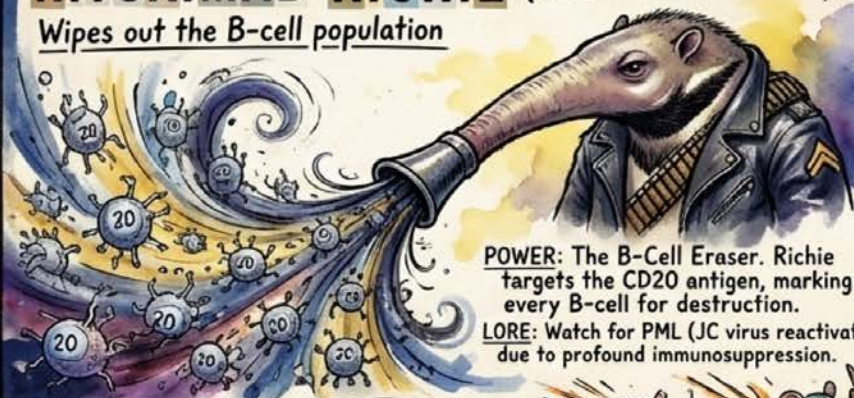


POWER: The Cloak Ripper. Penny rips the "Invisibility Cloak" off so the T-cells can attack.

LORE: Unleashes the immune system, but can cause severe autoimmune toxicities.

RITUXIMAB RICHIE (The CD20 Eraser)

Wipes out the B-cell population



POWER: The B-Cell Eraser. Richie targets the CD20 antigen, marking every B-cell for destruction.

LORE: Watch for PML (JC virus reactivation) due to profound immunosuppression.

BLEOMYCIN BEN (The Lung Scar-er)

Pulmonary Fibrosis



POWER: Oxidative Shredding. Ben generates free radicals that shatter DNA into pieces.

LORE: Causes devastating, irreversible pulmonary fibrosis.

KayfaBOGGLE!
Stage it, grade it!

NO FUTURE
FOR TUMORS!

TAEMO
REENTRIST?

NO FUTURE
FOR TUMORS!

ANARCHY IN
THE VEINS!

CHEMO
RIOT!

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TAMOXIFEN TAMMY

(The Estrogen Switch)

Antagonist (Breast) /
Agonist (Bone/Uterus)



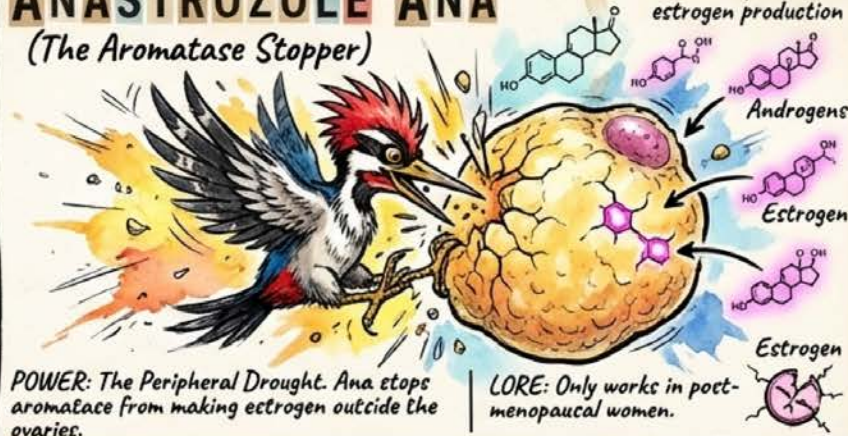
POWER: The Tissue Toggle. Tammy blocks estrogen in the breast, but acts like super-estrogen in the bone and uterus.

LORE: Risk of endometrial cancer!

ANASTROZOLE ANA

(The Aromatase Stopper)

Blocks peripheral
estrogen production



POWER: The Peripheral Drought. Ana stops aromatase from making estrogen outside the ovaries.

LORE: Only works in post-menopausal women.

ETOPOSIDE ED

(The Topo-II Tangler)

DNA supercoiling blocker



POWER: The Untwist Snap. Ed makes the DNA strands get stuck when they try to untwist.

LORE: The teneion bailde until the DNA completely chatters.

THE ONCOLOGY ATTENDING

(Master Synthesis)

Stage > Grade



POWER: The Final Word.

LORE: "Stage it, grade it, name the mutation, and don't kill the patient with the chemo. Stay Fluffy — and always double-check the labs!"

Doc FrizzleBob

KayfaBIZARRO!
Game Over.

KayfaBOGGLE!

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